

Does it spell Hindi right?

A focused evaluation of AI-generated posters for India's local shops, and a fair system for catching low-quality transcribers.

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INSIDE THIS DOCUMENT

One-page report

→

Q1 · The evaluation and why it matters

→

Q1 · How it works and how people judged

→

Q1 · Results and findings

→

Q1 · The eval as a product, and scale

→

Q2 · Spotting low-quality transcribers

→

Reflection, video script, appendix

→

In 30 seconds

Image models make beautiful posters. For a shop in Kanpur that wants its offer in Hindi, the question is simpler: **is the Hindi spelled right, and would the owner post it?** I tested this with matched English and Hindi versions of the same poster, so each model's "Hindi tax" can be measured directly.

MODELS

GPT Image 1

Gemini 2.5 Flash Image

Gemini 3.1 Flash Image Preview

LABELS USED

REAL DATA

MY AUDIT

ILLUSTRATIVE

PENDING

VIDEO WALKTHROUGH

[PASTE VIDEO LINK - unlisted YouTube / Google Drive, ≤2 min]

Hindi posters for local shops: how much worse do AI models get when the text is in Hindi?

THE EVAL

Category: marketing for India's micro-businesses. **Use case:** a shop owner in a Hindi-speaking town asks an AI tool for a WhatsApp Status poster (festival greeting, discount, price, notice) with exact Hindi text.

WHY IT MATTERS

India has 7.3 crore registered MSMEs, almost all micro, and WhatsApp is their shop window. Hindi is the first language of 52.8 crore people. One wrong matra makes a shop look careless, and a wrong word on a price or health notice misleads customers. Public image leaderboards rank overall taste on English prompts, so they can't see this failure.

SETUP

16 prompts across 8 real business types. **6 matched pairs** use the identical prompt in English and Hindi, which isolates the **Hindi tax**. 4 stress tests cover mixed scripts, long copy, dense conjuncts and Chhath Puja. One image per prompt per model, first output kept.

PARTICIPANTS

8 to 10 Hindi-reading adults (18+, consented) rate all 48 posters blind, in random order, on a phone app I built: text accuracy, cultural fit, "would you post it?" and best of three. **Main metric: ship-ready rate**, the share rated "post as-is".

KEY FINDINGS

- **English is solved; Hindi splits the models.** All three spelled English perfectly (6 of 6). On the identical Hindi posters: Gemini 3.1 Flash Image Preview 6 of 6, GPT Image 1 2 of 6, Gemini 2.5 Flash Image 0 of 6. [MY AUDIT](#)
- **One model generation closed the gap.** Gemini 2.5 → 3.1 went from 0 to 9 of 10 Hindi posters exactly right, so India-specific results go stale fast. [MY AUDIT](#)
- **Looks great ≠ usable.** Every poster looked professional, yet Gemini 2.5's chemist notice got every word wrong, including the day of the free health check. Unrequested text and fake brands appeared across models. [MY AUDIT](#)
- **PENDING** Participant ship-ready ratings fill in from the rating files.

Most important takeaway. "Looks great" and "usable" split apart exactly where India is different: the script. Labs should track **ship-ready rate per script**, not overall image quality. Products should never trust an image model to spell Indic text unchecked: verify it with OCR, or typeset the text on top of a generated background.

Q2 in one line. Most of the `user_id` column turned out to be task IDs, so I validated signals against a hidden gold set (41 clips, ~126 transcribers each). Tasks submitted faster than the audio's own length were wrong 99.6% of the time vs 26.5% otherwise. My system combines independent signals with unannounced test clips, and never blocks anyone without a human listening to the audio.

The evaluation and why it matters

What I tested, for whom, and why this was worth testing over nine other ideas.

What I evaluated

THE USER

The owner of a small shop in a Hindi-speaking Tier-2 or Tier-3 town: a mithai shop in Kanpur, a kirana in Indore, a tailor in Jaipur. They market on WhatsApp Status and groups, can't afford a designer, and need a new poster for every festival and offer.

THEIR JOB TO BE DONE

"Give me a poster I can post right now, in my customers' language, with my exact words and price."

WHAT BREAKS

Image models handle English text well but often produce Hindi that is almost right: a dropped nukta, a misplaced ि matra, a broken conjunct, or fake letters that only look like Devanagari. To someone who doesn't read Hindi, the poster looks perfect. To the shop's customers it looks careless.

WHAT "BETTER" MEANS

A poster the owner would **post without edits**. In order: (1) exactly the requested text, (2) nothing unrequested or misleading, such as extra text or fake brands, (3) visuals that fit the shop and the occasion, (4) looks good.

Why this eval, and not the others

I listed ten India-specific ideas and scored each 1 to 5 on the brief's criteria: India-specificity, business use, whether it exposes model differences, ease of prompting and judging, scale, originality, value to a lab and noise. The deciding question was: **can 8 to 10 ordinary raters judge it reliably, and does the answer drive a product decision?**

IDEA	SCORE /50	VERDICT
Hindi-text posters for local shops	48	Chosen. Very Indian, very common, and checkable: any Hindi reader can compare a spelling with the requested text
Health awareness posters in Indian languages	41	Same core skill, rarer use. Folded in as one pair (a chemist's free sugar test notice)
Food-delivery dish photos	37	Judging a regional dish depends on where the rater is from
Wedding invitation designs	37	Overlaps with posters, and is religiously sensitive
Ethnic-wear catalogue shots	36	Big market, but judging a saree drape needs experts, not 8 lay raters
Regional festival imagery	33	Subjective with no clear decision. Folded in as one Chhath Puja prompt
Hindi-labelled school diagrams	33	Errors are mostly generic (labels, science), not Indian
Indian faces and skin tones	32	Already well studied
Rural/urban stereotype audit	31	Important but noisy, and no clear product decision
Farming advisory illustrations	29	Needs agronomists to judge

Why it matters for India, and why an AI lab should care

- **The users are many.** India has over 7.3 crore Udyam-registered enterprises, nearly all micro ([MSME Year-End Review 2025](#)). India is WhatsApp's largest market, with 500M+ users ([TechCrunch](#)).
- **The language is big and the script is hard.** Hindi is the first language of 52.8 crore people ([Census 2011](#)), and Devanagari also serves Marathi, Nepali and Konkani. Vowel signs reorder (ि is written before the consonant it follows), consonants fuse into conjuncts (क्ष, ज्ञ, श्र), and small dots change meaning. A model can be excellent at English typography and still fail here.
- **Generic benchmarks miss it.** Arenas like the ones in the brief rank overall preference, mostly on English prompts, by voters who aren't checking Hindi spelling. A model can top them and still be unusable for this user.
- **The result drives a decision.** For a lab: is Indic text a weakness, and which words break (which tells you what training data to add)? For a product team building a "make my poster" feature: which model to use for Hindi, and **whether a text-checking layer is required before launch**.



How the evaluation works

Prompt design, models, fairness rules, and how participants judged the outputs.

The prompt set

Every prompt comes from **one fixed template**, so only the shop, the scene and the text change (all 16 are in Appendix A):

Design a square promotional poster for WhatsApp Status for {business}. {scene}. The poster must contain ONLY the following text, in English / in Hindi (Devanagari script), spelled exactly as given: Line 1: "..." Line 2: "..." Do not add any other words, letters, logos or watermarks. Style: bright, clean, professional design that a small local shop in India would proudly share with its customers.

6 MATCHED PAIRS • THE HINDI TAX	4 STRESS TESTS
Same prompt and scene, text in English or Hindi: • Diwali greeting · mithai shop • 10% off · kirana store • ₹250 blouse stitching · tailor • Admissions, 3 lines + phone · coaching centre • Holi herbal colours · general store • Free sugar test · chemist	• Mixed scripts: English + Hindi on one poster (mobile shop) • Long text: 4 lines, ~17 words (tiffin service) • Dense conjuncts: श्री कृष्ण जन्माष्टमी (sweet shop) • Less-represented festival: Chhath Puja (Patna fruit shop) Difficulty climbs from a short greeting to long text. Real traps are spread throughout: nukta (सिर्फ़), chandrabindu (जाँच), conjuncts (क्ष, झ), ₹, digits and a phone number. Every line is something a real shop would write.

Using the same template, quoting the exact text and saying "ONLY this text" means differences come from the models, not from vague prompts. The English versions act as a control for prompt difficulty.

Models and fairness

COMPANY	MODEL	WHERE GENERATED	SETTINGS	STATUS
OpenAI	GPT Image 1 (gpt-image-1)	OpenAI Image API	1024×1024, quality high, 1 image	16 of 16 generated, 2026-10-07
Google	Gemini 2.5 Flash Image (gemini-2.5-flash-image)	Google Gemini API	1:1, default 1K	16 of 16 generated, 2026-10-07
Google	Gemini 3.1 Flash Image Preview (gemini-3.1-flash-image-preview)	Google Gemini API	1:1, default 1K	16 of 16 generated, 2026-10-07

Fairness rules: identical prompt text and square format for every model. **One image per prompt per model, and the first output is kept** (the script won't overwrite, so there is no cherry-picking). Refusals count as "would not post". Every image is logged with its timestamp and settings (Appendix D). GPT Image 1 has a quality setting (I used high, as a shop owner

would); the Gemini models don't, so they use their default. I didn't add a fourth model, because it would add a third more rating work without changing the decision.

How participants judged

Poster Eval

4 / 17

Shop: Laxmi Tailors
Requested text (must match exactly):
Blouse Stitching Only ₹250
Laxmi Tailors

Blouse Stitching
Only ₹250



Laxmi Tailors

1. Text accuracy

Exactly correct

Small error

Major error

Gibberish / missing

2. Looks right for this shop & occasion?

Looks right

Something is a bit off

Clearly wrong / not Indian

wrong festival / ritual items

not Indian-looking setting or people

wrong food / product

Distorted objects, hands or faces

fake or misspelled brand / logo

extra unrequested text

3. Would you post it?

Post as-is

Post after small fixes

Would not post

Blouse Stitching
Only ₹250



Laxmi Tailors

1. Text accuracy

Exactly correct

Small error

Major error

Gibberish / missing

2. Looks right for this shop & occasion?

Looks right

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Clearly wrong / not Indian

wrong festival / ritual items

not Indian-looking setting or people

wrong food / product

Distorted objects, hands or faces

fake or misspelled brand / logo

extra unrequested text

3. Would you post it?

Post as-is

Post after small fixes

Would not post

Blouse
Stitching
Only ₹250



लक्ष्मी टीलर्स

1. Text accuracy

Exactly correct

Small error

Major error

Gibberish / missing

2. Looks right for this shop & occasion?

Looks right

Something is a bit off

Clearly wrong / not Indian

wrong festival / ritual items

not Indian-looking setting or people

wrong food / product

Distorted objects, hands or faces

fake or misspelled brand / logo

extra unrequested text

3. Would you post it?

Post as-is

Post after small fixes

Would not post

Which ONE would you post for this shop?

None of these

A

B

C

Anything specific you noticed? (optional)

The rating app (real, working). The requested text is pinned at the top; three blinded posters sit side by side; three questions per poster, then "which one would you post?".

THE QUESTIONS 1. Text accuracy: exactly correct · small error (1 or 2 letters off, still readable) · major error (wrong, missing or extra words) · gibberish or missing 2. Looks right for this shop and occasion? yes · a bit off · clearly wrong, plus tags such as <i>fake brand</i> or <i>extra text</i> 3. Would you post it? as-is · after small fixes · no 4. Which one would you post? A · B · C · none	WHY THIS METHOD With only 8 to 10 people, every minute of rating must count. Pure A/B voting gives one bit per vote and hides <i>why</i> a model lost. Rating each poster against the exact text is close to objective. Seeing all three side by side keeps judgments consistent, and the final pick gives preference data as well.
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RISK	WHAT I DID ABOUT IT
Knowing the model biases the rating	Images carry random IDs; the key never ships in the app
Position or order effects	Poster positions and prompt order are shuffled for each person
Misunderstanding the scale	A practice poster with one wrong matra, with feedback
Careless or inconsistent rating	One screen is repeated at the end; people who average under 15 s per screen are excluded
Rating looks, not text	The text question comes first; the reference text is pinned; tap to zoom
Eligibility	18+ gate, consent checkbox, "can you read Hindi?" check

Design in numbers: 16 prompts $\times$ 3 models = 48 posters. Every participant rates every poster (16 screens + 1 repeat, about 20 minutes). With 9 people that is 432 poster ratings and 144 best-of-three picks.

What this sample can and can't prove. Ratings of the same poster aren't independent, so the real sample size is **16 prompts**, not 432 ratings. I use intervals that resample both prompts and raters. With 16 prompts, only gaps of **about 25 percentage points or more** count as a real direction; smaller gaps are reported as description. Agreement between raters is measured with Krippendorff's alpha. I planned this analysis before seeing any ratings, and I deliberately avoid significance tests on single prompts.



Results and findings

What I can show today, clearly labelled by where it comes from.

Results

My letter-by-letter audit of the outputs MY AUDIT

Before the study, I read every poster against the requested text. This is my own inspection of real outputs, not participant data; it will be compared with the participants' labels.

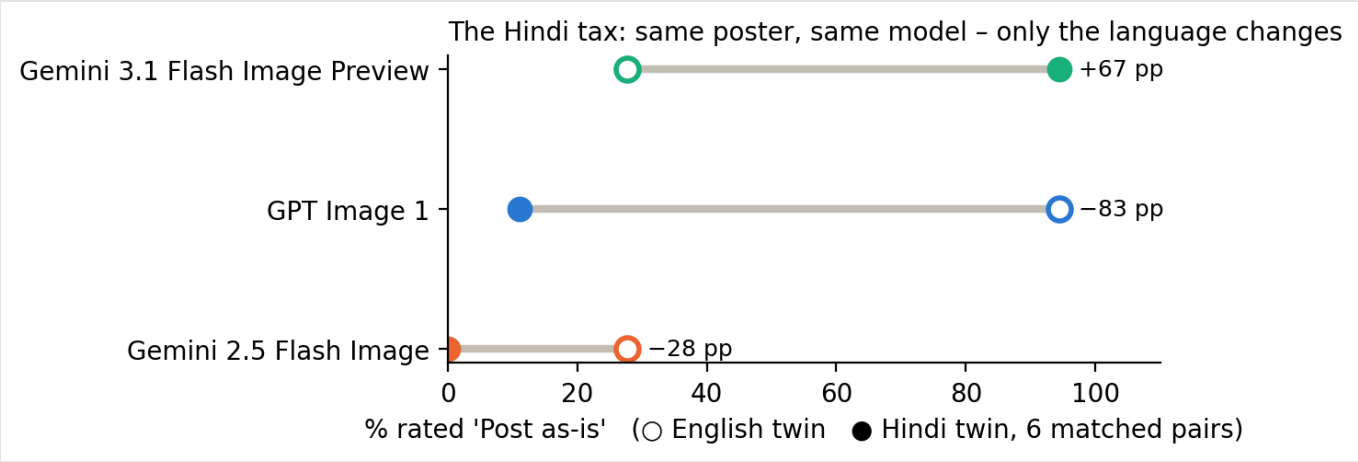
MODEL	ENGLISH TEXT EXACT	HINDI TEXT EXACT (ALL 10 HINDI/MIXED)	HINDI EXACT - 6 TWINS ONLY	HINDI READABLE (EXACT + SMALL)	HINDI MAJOR/GIBBERISH	UNREQUESTED TEXT	FAKE BRAND/LOGO
GPT Image 1	6/6	3/10	2/6	7/10	3/10	4/16	2/16
Gemini 2.5 Flash Image	6/6	0/10	0/6	0/10	10/10	5/16	3/16
Gemini 3.1 Flash Image Preview	6/6	9/10	6/6	10/10	0/10	1/16	1/16

Per-image notes are in Appendix C.

AI-judge panel AI, NOT HUMAN

To get an early read and to test the "can we automate judging?" question from the scaling plan, I ran the exact participant protocol with an AI judge: Claude Opus 5 saw the same blinded, shuffled screens and answered the same questions, in 3 independent runs. **This is not participant data and does not replace the human study.**

MODEL	WOULD POST AS-IS (95% CI)	TEXT EXACTLY CORRECT	PICKED AS BEST	HINDI TAX (POST AS-IS)
Gemini 3.1 Flash Image Preview	65% (44% to 85%)	94%	65%	-67 pp (28% → 94%)
GPT Image 1	46% (21% to 71%)	44%	29%	83 pp (94% → 11%)
Gemini 2.5 Flash Image	10% (0% to 25%)	31%	6%	28 pp (28% → 0%)



How far can it be trusted? Agreement between the three AI runs on text accuracy: Krippendorff's alpha 0.95. Its median text label matched my letter-by-letter audit on **85%** of posters (98% within one level). Same label on the repeated screen: 100%. The judge also transcribes what it reads, which makes every label auditable.

Where it diverges, and why that matters. On "would you post it?" the AI judge is very strict about tiny background lettering in photo-realistic scenes: garbled text on packet labels, book spines or a clock face. It downgraded Gemini 3.1's English posters for this even when the headline text was perfect, which produces an odd negative "Hindi tax" for that model. A shop owner may never notice those details. So the AI judge is ready to pre-screen **text accuracy**, but **"would you post it?" should stay a human judgment** until it is checked against the participant ratings.

Participant study (humans)

PENDING

Participant ratings. This section fills in automatically from the rating files: the leaderboard with 95% intervals, the Hindi tax per model, the per-prompt view, rater agreement, repeat-screen consistency and position bias. Nothing here is estimated or simulated.

Findings

These findings come from my letter-by-letter audit of all 48 real outputs. The participant ratings will confirm or overturn them.

F1 · English is solved. Hindi separates the models completely		MY AUDIT
OBSERVATION	All three models got the English text exactly right on 6 of 6 posters. On the identical Hindi versions: Gemini 3.1 Flash Image Preview 6 of 6, GPT Image 1 2 of 6, Gemini 2.5 Flash Image 0 of 6.	
EVIDENCE	Hindi tax on exact text: 0 pp, 67 pp and 100 pp . Across all 10 Hindi prompts: 9, 3 and 0 exactly right. Gemini 2.5 produced 4 unreadable posters; Gemini 3.1 produced none.	
INTERPRETATION	The pairs share everything except the language, so this gap is pure script rendering. An English-only benchmark would score these three models as roughly equal.	
IMPLICATION	A "make my poster" feature tested in English could ship to Hindi users with a model that cannot spell a single Hindi poster correctly.	
RECOMMENDATION	Measure and gate launches per script. On this sample, Hindi posters should go to Gemini 3.1, with an OCR check as a safety net.	

F2 · One model generation closed the gap		MY AUDIT
OBSERVATION	Within the same company, Gemini 2.5 Flash Image → Gemini 3.1 Flash Image Preview went from 0 of 10 to 9 of 10 exactly right in Hindi, while English stayed perfect.	
INTERPRETATION	India-specific quality moves fast and independently of English quality. A lab that tested last year's model would wrongly conclude that Hindi posters can't ship.	
RECOMMENDATION	Treat this as a regression suite, not a one-off study: version every leaderboard entry and re-run the frozen prompts on every model release (the admin dashboard's "model version changed" alert).	
F3 · Looking great is not the same as being usable		MY AUDIT
OBSERVATION	All 48 posters look professional, with correct festival objects and Indian settings. Yet Gemini 2.5's chemist notice reads "हर रनिवार मुफ्टि शुमार जॉच्छ / श्री मडिजल स्तोष": every word is wrong, including the day of the free health check.	
INTERPRETATION	A preference vote by someone who doesn't read Hindi would rate this poster highly. For the shop, it is unusable, and on a health notice it is misleading.	
RECOMMENDATION	India-focused evals need native-script raters and a "would you post it?" question, not just "which looks better?".	
F4 · Same mistake, very different consequence		MY AUDIT
OBSERVATION	Despite "ONLY the following text", models added text nobody asked for: GPT Image 1 on 4 of 16 posters, Gemini 2.5 on 5, Gemini 3.1 on 1.	
EVIDENCE	In English the extra shop name was readable (GPT, P09). In Hindi it became fake words: GPT's "अगरावी गसरलातो सोरर", Gemini 2.5's "मां तारा फ़ोट बन्दार, पतना, बेहार". Gemini 3.1 printed the instruction word "WhatsApp Status" as a heading.	
RECOMMENDATION	Measure instruction-following per script; in product, strip or OCR-check any text that wasn't requested.	
F5 · Culture is not the bottleneck. Text and brands are		MY AUDIT
OBSERVATION	All three models got the visual culture right, even for Chhath Puja: soop with thekua, sugarcane, women offering arghya at a river ghat. Gujiya and pichkari for Holi, a brass thali of mithai for Diwali.	
EVIDENCE	Brands appeared on 6 of 48 posters: fake ones ("AASHIRAAD", "GUPIA ATTA") and real ones (Apple, USHA).	
INTERPRETATION	For this use case the frontier is spelling and restraint, not cultural knowledge. Fake or real brand logos put a trademark risk on a shop owner who won't notice.	
RECOMMENDATION	Block logos in business templates; keep "fake brand" as a rating tag (already in the app).	

An exploratory note: GPT Image 1's errors clustered in less common words (मोबाइल → मोबाल, डिलीवरी → डिलेबरी) while its densest conjuncts were perfect, which suggests memorised greetings. Gemini 2.5 failed on common and rare words alike, so this pattern is model-specific. I noticed it after seeing the outputs, so it is a hypothesis, not a finding.

What an AI lab or product team should do

1. **Measure per script, ship per script.** One overall quality score hides the Hindi tax. Set a launch bar such as "80% ship-ready in Hindi" before a poster feature goes live in Hindi.
2. **Don't let the image model own the text.** Generate a background without text, then typeset the exact words with a proper Indic font engine. If the model keeps rendering text, read it back with OCR, compare it with the request, and retry or fall back on a mismatch.
3. **Route by script.** If one model wins on Hindi text and another on visuals, send Hindi requests to the first.
4. **Add guardrails that matter here:** strip text nobody asked for, block brand logos in business templates, and take extra care with price and health notices.
5. **Fix the data, not just the model.** Errors cluster in less common words, English loanwords written in Devanagari, and nukta. A targeted training set of shop names, prices and loanwords is cheaper than more general data.



The eval as a product, and how it scales

The smallest product that is useful: a rating app, an honest leaderboard and a failure explorer.

WHO	WHAT THEY DO	SCREEN
Rater (Hindi reader, 18+)	Opens a link, consents, practises, rates 17 screens	Rating app (built, shown above)
Eval team at Josh Talks	Launches a run, watches quality, handles alerts	Admin dashboard
Model or product team	"Which model for Hindi posters?" then digs into failures	Leaderboard, failure explorer

Leaderboard

Ranked by **ship-ready rate in Hindi**, with 95% intervals. **Models whose intervals overlap share a rank**, because the data can't separate them. The English score and the Hindi tax sit alongside, so the gap is visible. A model is marked *provisional* until it has 300+ ratings. Filters: script, business type, difficulty, rater state, eval run.

Bharat Poster Arena

Leaderboard

Explore failures

Compare versions

Methodology

ILLUSTRATIVE NUMBERS • LAYOUT ONLY

Script: Devanagari (Hindi) • Business: All 8 • Difficulty: All • Rater pool: Hindi-native, 18+ • Eval run: 2026-10 • v1 •

Which model makes posters a local Indian shop would post as-is?

Ranks populate from data. Ranked by **Ship-ready rate** (% of ratings "Post as-is") on Hindi prompts. Models whose 95% intervals overlap share a rank band - we do not split hairs the data cannot support.

RANK	MODEL	SHIP-READY (HINDI)	95% INTERVAL	ENGLISH TWIN	HINDI TAX	TEXT EXACT	LOOKS INDIAN	RATINGS	STATUS
-	Gemini 3.1 Flash Image Preview Google · gemini-3.1-flash-image-preview	xx%		xx%	-xx pp	xx%	xx%	432	stable
-	GPT Image 1 OpenAI · gpt-image-1 · quality=high	xx%		xx%	-xx pp	xx%	xx%	432	stable
-	Gemini 2.5 Flash Image Google · gemini-2.5-flash-image	xx%		xx%	-xx pp	xx%	xx%	432	stable

Hindi tax = Ship-ready(English twin) - Ship-ready(Hindi twin) on matched prompt pairs. Interval = two-level bootstrap over prompts and raters. "Provisional" = <300 ratings or rater agreement $\alpha < 0.4$ for that model.

Where each model breaks (Hindi)

Ship-ready rate by difficulty level

LEVEL	GPT IMAGE 1	GEMINI 2.5	GEMINI 3.1
L1 short greeting	xx	xx	xx
L2 offer / price	xx	xx	xx
L3 multi-line / code-mixed	xx	xx	xx
L4 long text	xx	xx	xx

Why posters get rejected

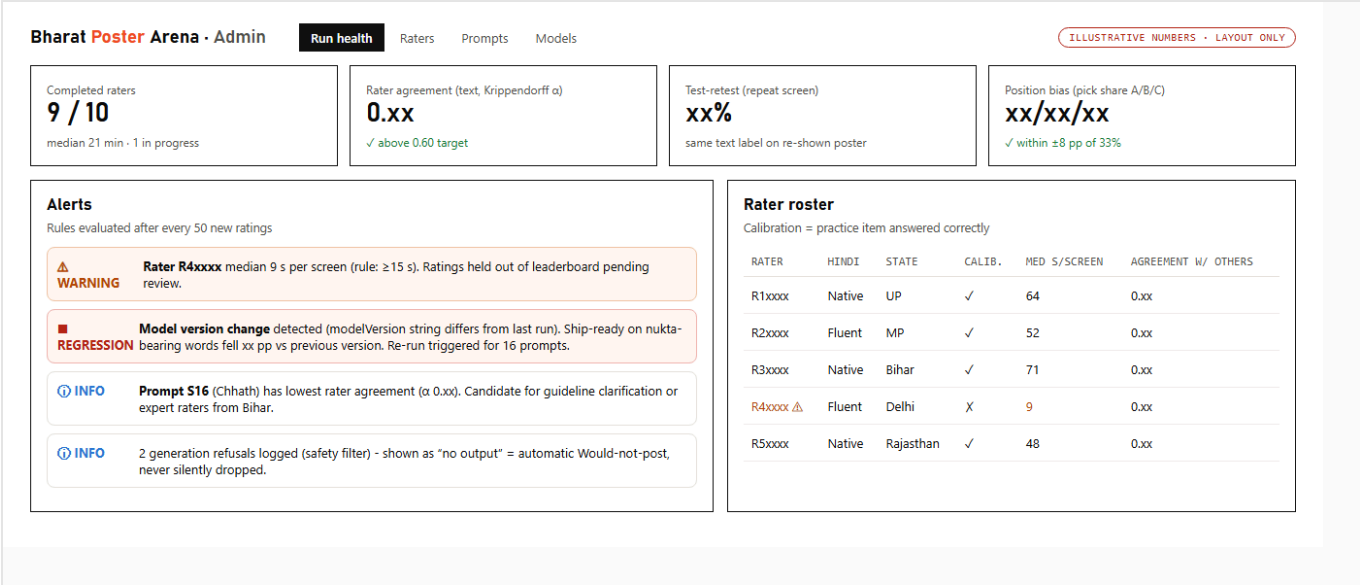
Share of "Would not post" ratings, by reason tag

REASON	GPT 1	GEM 2.5	GEM 3.1
Misspelled / gibberish Hindi	xx%	xx%	xx%
Extra unrequested text	xx%	xx%	xx%
Fake brand / logo	xx%	xx%	xx%
Wrong festival items	xx%	xx%	xx%

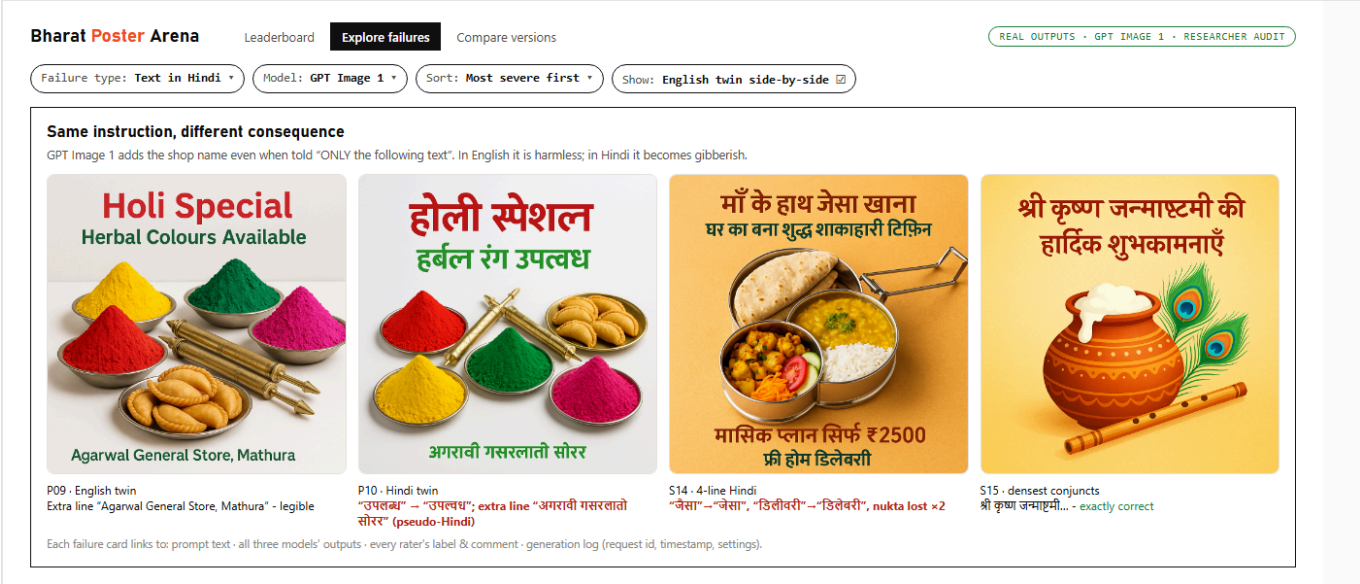
Leaderboard mockup. Numbers are placeholders ("xx"); ranks fill in from data.

Admin dashboard and failure explorer

The admin view tracks completion, rater agreement, repeat-screen consistency and position bias. Alerts fire for raters who rush, for low-agreement prompts, and **when a provider's model version changes** (which re-runs the frozen prompt set). The failure explorer is where teams learn the most: filter by failure type, see the English twin next to the Hindi one, and open every rater's comment.



Admin dashboard mockup (illustrative numbers).



Failure explorer. Thumbnails are real outputs with my audit notes.

Scaling plan

STAGE	RATERS	WHAT GROWS	QUALITY CONTROL	AUTOMATION
Pilot (now)	8 to 10	16 prompts, Hindi, 3 models	practice item, repeat screen, rater agreement	none
Next	~50	60 prompts, 2 images per prompt (separates the model from luck), add Marathi	hidden test posters with known errors; drop raters below 80% on them	OCR pre-checks text; humans confirm
Then	~500	8 to 10 scripts (Bengali, Tamil, Telugu, Gujarati, Gurmukhi, Urdu...), more business types, new models monthly	per-script agreement targets, adjudication, state and gender balance	extra raters only where OCR is unsure or raters disagree
Platform	thousands	live leaderboard, private evals for labs	rater reputation, drift checks, fixed anchor prompts in every run	automatic re-run when a model version changes

Where raters come from: Josh Talks already has 20,000+ daily transcribers who read Indian languages. Q1's rating platform and Q2's quality system are the same machine: people whose work is trusted *because* it is measured.

What stays human: "would I post this?", cultural fit and anything sensitive. **What can be automated:** exact text checking with OCR and string matching, extra-text and logo detection. One caution: a vision-LLM judge tends to read a misspelled word as the word it should be, so any automated judge must be checked against human labels before it is trusted.

Spotting low-quality transcribers

Part I finds the warning signs in the data. Part II turns them into a system that is fair enough to block accounts.

How the work happens, and where it can be gamed

Audio → Whisper draft → worker listens → worker edits → submit → paid per accurate hour.

SHORTCUT	WHAT IT LOOKS LIKE IN THE DATA
Not listening	submitted faster than the audio plays
Accepting the Whisper draft blindly	user text = Whisper text, even when Whisper is wrong
Cosmetic edits to look busy	only spaces or punctuation change
Placeholders	[blank] , ".", "...." on clips that have speech
Accepting Whisper's made-up text	known fake strings like "प्रस्तुति प्रस्तुति" submitted as-is

What the data really contains REAL DATA

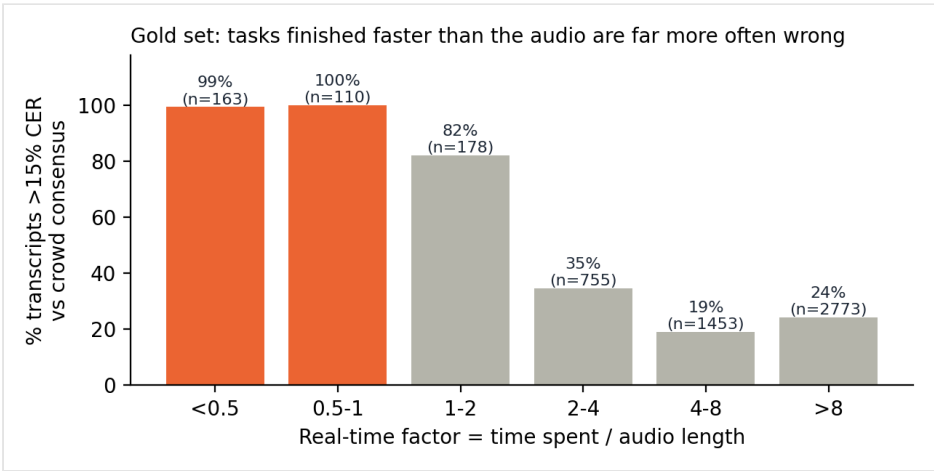
Before setting thresholds I checked the file (51,783 rows). Four findings changed the answer:

10 OF 48,875 USER_IDS ARE REAL PEOPLE. THE REST ARE TASK IDS	99.4% OF ROWS: "CHARS PER SECOND" DIVIDES BY AUDIO LENGTH, NOT TIME TAKEN	4 to 12 s THE 10 USERS' TIMES: WHOLE NUMBERS, UNRELATED TO CLIP LENGTH	41 × 126 HIDDEN GOLD SET: CLIPS × MEDIAN TRANSCRIBERS EACH
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- **Only 10 IDs repeat** (86 to 609 tasks each). The other 48,865 appear once, and consecutive IDs share a recording folder 83% of the time, so they are task IDs. Per-person analysis is only possible for 10 users.
- **segment_character_per_second is mislabelled.** It equals characters ÷ *audio duration* (99.4% of rows), not ÷ time taken (0.07%). It measures how dense the transcript is, not how fast the worker typed.
- **The 10 users' timings look generated.** Always whole numbers, spread evenly from 4 to 12 s, with no link to clip length (correlation −0.01, vs 0.54 for everyone else). I make no time-based judgment about these users.
- **is_edited is inflated.** 83% say "Yes", but 23,104 of those rows have no Whisper text at all, and 963 differ only in spacing or punctuation.
- **A hidden gold set:** 41 test clips, each transcribed independently by a median of 126 people. I used the most central transcription as the reference answer and marked any task with over 15% character error as bad. That let me **test** which signals predict bad work instead of guessing.

Part I · Warning signs

Warning sign 1 · Listening less than the audio		STRONGEST
WHAT IT TELLS US	You can't transcribe a 10-second clip you didn't hear. The minimum honest time is one full listen.	
HOW TO MEASURE	Speed ratio = time taken ÷ audio duration , per task, on clips of 2 s or more. Per user: share of the last 100 such tasks below 1.	
EVIDENCE	In the gold set, tasks below 1 were bad 99.6% of the time (273 tasks) vs 26.5% for the rest. 1 to 2: 82% bad; 2 to 4: 35%; 4+: about 20%. It also catches workers who <i>do</i> edit: edited tasks at 1 to 2 were still 70% bad. In normal production work, 18.7% of tasks were submitted faster than the audio , vs 4.9% on test clips. People listen when it might be a test.	
RED FLAGS	Below 0.5 is impossible even at 2× playback. Between 0.5 and 1 is only possible at fast playback. A user with 30%+ of tasks below 1 gets more QA; 20%+ below 0.5 can lead to restriction. Minimum 30 tasks first.	
FALSE ALARMS	Experts using 1.5 to 2× playback; clear audio where Whisper is already right; timer bugs on low-end phones or poor networks; very short clips. Fix: log playback speed and actual audio played, ignore clips under 2 s, act only on patterns across many tasks.	



Warning sign 2 · Accepting Whisper without checking

STRONG

WHAT IT TELLS US

The job is to *fix* Whisper. Submitting its draft unchanged when Whisper is wrong means the worker didn't really listen.

HOW TO MEASURE

Normalise both texts (Unicode form, remove punctuation and extra spaces, so cosmetic edits don't count), then check whether they are equal. Per user: unedited share on clips of 5 s or more, with a 95% interval. **Better: hidden test tasks**, real clips whose Whisper draft has a known, verified error, mixed in at about 1 in 25.

EVIDENCE

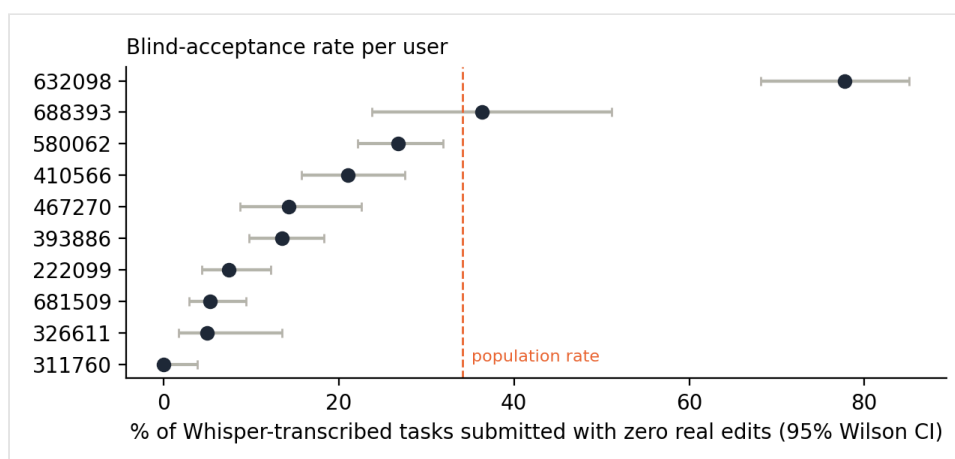
Whisper was wrong on every gold clip, yet 8.6% of submissions accepted it untouched. Across production, the unedited rate is 34%. **User 632098 left Whisper untouched on 78% of tasks (68 to 85%)**, 84% on longer clips, including obvious errors such as "लाओं से ... सहर में" for "गाँव से ... शहर में".

RED FLAGS

Lower end of the user's interval above 60% (30+ tasks) → more QA. Hidden tests: 3 of the last 10 accepted → restrict; 5 of 10 → human review. A careful worker who misses 10% of planted errors hits 5 of 10 only **0.16%** of the time.

FALSE ALARMS

Whisper genuinely right on clear speech; very short clips; a worker only on easy audio. Fix: measure on longer clips, compare with peers on the same language and audio quality, and rely on hidden tests for anything serious.



More signals (use together, never alone)

SIGNAL	EVIDENCE	USE
Accepting Whisper's made-up text. Whisper repeats fake strings on near-silent clips	"प्रस्तुति प्रस्तुति" appears on 398 clips: 62% of people wrote [blank], 17% submitted it as-is. User 580062 accepted 4 of 12	3+ acceptances and 50%+ of encounters → more QA
Missing content. Text much shorter than expected	Under half the reference length: all 205 such gold tasks were bad	More QA, stronger with signal 1
Placeholder on speech. [blank] where Whisper heard real words	All 11 gold cases were bad	Nudge, then QA if repeated
Junk entries. Only punctuation, links, wrong script	User 393886 typed "." on 39 near-silent clips	Block in the app with a helpful message. A guideline problem, not fraud

Weak signals I would not use alone: the `is_edited` flag (inflated); raw time taken (depends on clip length); characters per second (fast speakers and tags on tiny clips give extreme values); overall blank rate (19% of clips really are silent); very

long times (idle tabs); repeated sounds like "ह ह ह" (real laughter; my first junk rule flagged these, so I removed it).

The 10 users today

Text signals only, since their timings can't be trusted:

USER	TASKS	UNTOUCHED WHISPER (95% CI)	OTHER EVIDENCE	ACTION
632098	165	78% (68 to 85%)	84% on longer clips; obvious errors left in	Human review of 20 recent long clips against the audio; payouts wait for QA. Not an automatic block
688393	86	36% (24 to 51%)	48% on longer clips; small sample	3× more QA
580062	609	27% (22 to 32%)	accepted 4 of 12 fake strings; 21 punctuation-only entries	More QA and a guideline nudge
410566	349	21% (16 to 28%)	2 of 5 fake strings accepted	Nudge
Other 6	149 to 509	0 to 14%	nothing serious	No action

Part II · The detection system

Principle: behaviour (speed, edit patterns) only decides *where to look*. Serious actions need proof of **wrong output**: hidden test tasks, or QA on the real audio. No single metric ever blocks anyone.

LEVEL	TRIGGER (OVER A ROLLING WINDOW, MINIMUM EVIDENCE)	ACTION	WAY BACK
0 · Prevent	always on	App rejects punctuation-only or wrong-script submissions, requires the audio to be played, offers one [blank] button	n/a
1 · Nudge	one signal at a soft level, 30+ tasks	In-app tip: "6 long clips were submitted faster than their audio. Please listen fully." No pay impact	clears after 7 clean days
2 · Look closer	one strong signal, or two soft ones, 50+ tasks	QA on 15% of tasks; payouts for those wait for QA	QA error ≤ 5% over 20 audits
3 · Restrict	two independent signals strong, or QA error above 10%	Easier tasks only, daily cap, all work QA'd, clear notice	2 weeks at QA error ≤ 5%
4 · Human review	level 3 for 14 days, 5 of 10 hidden tests failed, or clear fraud (shared device, scripted timing)	A trained reviewer listens to 20+ recent tasks; the worker can explain	reviewer decision
5 · Block	all of: reviewer confirms error far above target on real audio, two or more independent signals, and a prior warning with 7 days to improve (waived only for proven fraud)	Block, pay for verified good work, appeal within 14 days to a different reviewer	appeal

The rule in the brief's format. If a transcriber (X) keeps submitting faster than the audio could be heard, and/or (Y) leaves known Whisper errors uncorrected on hidden tests, and/or (Z) submits blanks or junk on clips with speech, then raise QA and restrict. **Block only when** at least two of X, Y and Z hold, **and** a reviewer confirms from the audio that their error rate is far above target, **and** they were warned and given time to improve.

If I were a bad transcriber, how would I game this?

TRICK	COUNTER
Leave the tab open to pad the time	Measure seconds of audio actually played, plus tab focus, not wall-clock time
Play the audio muted	Hidden tests: you can't fix an error you didn't hear
Add a space or full stop to look "edited"	Compare normalised text, where cosmetic edits don't count
Make random small edits	Hidden tests and QA measure correctness, not activity
Learn which tasks are tests	Tests use real production clips, look identical and rotate
Mark hard clips [blank]	Blank-on-speech signal; QA before paying for blanks
Stay just under the thresholds	Thresholds aren't published; independent signals; random QA for everyone

What engineering needs to build

METRIC	DATA NEEDED	WINDOW	THRESHOLD → ACTION	MONITOR
Audio coverage and speed ratio	play/pause/seek events, playback speed, focus, open and submit time	last 100 tasks ≥ 2 s, min 30	below 0.5 on 20%+ → level 3 candidate; below 1 on 30%+ → level 2	spread by language and device
Hidden test acceptance	test ID, planted error, submitted text	last 10 tests	3 of 10 → level 3; 5 of 10 → level 4	careful workers must pass 90%+
Untouched Whisper rate	Whisper text and version, user text	last 100 clips ≥ 5 s	interval lower end > 60% → level 2	baseline per language and Whisper version
Made-up text accepted	weekly auto-mined list of fake strings	90 days	3+ and 50%+ → level 2	list growth
QA error rate (the ground truth)	reviewer's corrected text	20+ audits	> 10% → level 3; confirmed in review → level 5 eligible	reviewer agreement on a 5% overlap

Also needed:

- **Fix the data first:** a real `user_id` on every task, a correct definition for the characters-per-second field, a look at why one group's times are whole numbers from 4 to 12, and playback speed logging.
- **Guardrails:** no action under 30 tasks; at most one level change per week; compare workers with peers on the same language, clip length and audio quality; re-baseline for 7 days whenever Whisper is upgraded.
- **Fairness:** workers see *what* was flagged and how to fix it (never the exact thresholds); every block has a named reviewer, an evidence bundle and an appeal.
- **Watch for false positives:** if more than 10% of appeals are overturned, blocking pauses automatically.

Rollout: (1) weeks 0 to 2: add logging, input checks and hidden tests, no enforcement. (2) Weeks 2 to 6: shadow mode, where signals are computed and QA confirms whether flagged workers really are worse, and thresholds are tuned so no more than 2% of workers reach level 2. (3) Weeks 6 to 10: nudges and extra QA only. (4) Then restrictions and review in one language first; blocking only once appeals are staffed. **Success:** lower error in accepted data, lower cost per accurate hour, an overturn rate under 10%, and **good workers staying**.

Final reflection

Short answers to the brief's five questions, and the script for the video walkthrough.

1. Why did you choose this eval? It is the narrowest idea I found that still answers a real product question. Hindi posters for local shops are common, commercially real and very Indian. With only 8 to 10 raters, they are also judgeable with little noise, since anyone who reads Hindi can check a spelling. The matched pairs turn "is it good for India?" into one number: the Hindi tax.

2. Why is it useful for India? Crores of small businesses sell on WhatsApp in their customers' language and can't afford designers. A tool that makes a beautiful poster with one wrong matra fails them without them noticing. This eval measures the thing that decides whether such a tool is usable here, and it extends to every Indian script.

3. Why would an AI lab building for India care? Global leaderboards can't see this failure, and neither can evaluators who don't read Hindi. The eval gives a per-script launch decision, a list of failure types that points to the training data needed, and a check to re-run on every new model version.

4. What did you learn from running the sample? That an English benchmark would have called these three models roughly equal, while in Hindi they ranged from perfect to unreadable. Checking every output before the study also changed the design: I added "fake brand" and "extra text" as rating tags because I saw them in real posters. And the biggest finding was one I didn't design for: a single model generation (Gemini 2.5 → 3.1) closed the Hindi gap, which makes the case for re-running this eval on every release rather than treating it as a one-off.

5. What would you improve with more time? Two images per prompt per model and about 60 prompts; raters balanced across Hindi-speaking states; a second Devanagari language (Marathi) and one other script (Tamil or Bengali); checking an OCR text scorer against human labels so people only judge what machines can't; and testing "background + typeset text" as a fourth option to measure the product fix directly. For Q2: run shadow mode on real platform logs with proper user IDs and playback events.

Video walkthrough script (under 2 minutes)

0:00 to 0:15 · Problem. "Picture a sweet-shop owner in Kanpur. Diwali's coming, so she asks an AI for a WhatsApp poster in Hindi. It looks gorgeous, but her shop's name is misspelled, and she can't tell. That's what I evaluated."

0:15 to 0:40 · What. "One narrow use case: Hindi posters for local Indian shops. I compared GPT Image 1, Gemini 2.5 Flash Image and Gemini 3.1 Flash Image Preview on 16 real posters: a kirana discount, a tailor's price, a coaching notice, a Chhath greeting. Six of them are twins: same prompt, only the language changes. So I can measure each model's Hindi tax directly."

0:40 to 1:10 · How. "8 to 10 Hindi readers rated all 48 posters blind in an app I built, with names hidden and positions shuffled. For each poster: is the text exactly right, does it look right for an Indian shop, and would you actually post it? That last one, the ship-ready rate, is my main metric."

1:10 to 1:45 · Findings. "Here's what surprised me. All three models spelled English perfectly. In Hindi, on the exact same posters? Gemini 3.1 got six out of six. GPT Image 1 got two. Gemini 2.5 got zero, and on the chemist's notice it even misspelled the day of the free sugar test. And every one of those posters looked professional, which is exactly the trap. Also, Google's newer model went from zero to nine out of ten in one generation, so a result like this goes stale fast."

1:45 to 2:00 · So what. "For labs building for India: measure ship-ready rate per script, not image quality. For products: never trust an image model to spell Indian scripts unchecked. Check it with OCR, or typeset it. And Q2 is in the doc: a fair way to catch bad transcribers. Thanks!"

Supporting material

Prompts, rating questions and consent, generation log, every generated image, and the Q2 tables. Code and full-size files: <https://github.com/AbhyudayPatel/josh-talks-hindi-poster-eval>

A · All prompts

ID	LANG · LEVEL	CATEGORY	EXACT TEXT REQUESTED	WHAT IT TESTS
P01 · pair A	EN · L1 short	Festival greeting (Diwali)	Happy Diwali / Sharma Sweets	Control for P02: same creative, Latin script
P02 · pair A	HI · L1 short	Festival greeting (Diwali)	शुभ दीपावली / शर्मा स्वीट्स	Shortest Hindi copy; reph conjunct (र्म), half-letter cluster (स्व, ट्स)
P03 · pair B	EN · L2 offer+number	Discount offer (kirana)	Flat 10% Off on All Groceries / Gupta Kirana Store	Control for P04
P04 · pair B	HI · L2 offer+number	Discount offer (kirana)	सभी किराना सामान पर 10% की छूट / गुप्ता किराना स्टोर	6-word offer line mixing Devanagari + digits + %; pre-posed i-matra (कि)
P05 · pair C	EN · L2 price (₹)	Service price (tailor)	Blouse Stitching Only ₹250 / Laxmi Tailors	Control for P06
P06 · pair C	HI · L2 price (₹)	Service price (tailor)	ब्लाउज़ सिलाई सिर्फ ₹250 / लक्ष्मी टेलर्स	Nukta (ज़, फ़), ₹ symbol, hard conjunct in a very common name (क्ष्मी)
P07 · pair D	EN · L3 multi-line	Admissions notice (coaching)	Admissions Open / Classes 9 to 12: Maths and Science / Call: 98765 43210	Control for P08
P08 · pair D	HI · L3 multi-line	Admissions notice (coaching)	प्रवेश प्रारंभ / कक्षा 9 से 12: गणित और विज्ञान / संपर्क करें: 98765 43210	3 lines, ~11 words; conjuncts प्र, क्ष, ज्ञ, क; 10-digit phone number
P09 · pair E	EN · L2 festival	Festival product (Holi)	Holi Special / Herbal Colours Available	Control for P10
P10 · pair E	HI · L2 festival	Festival product (Holi)	होली स्पेशल / हर्बल रंग उपलब्ध	English loanwords in Devanagari (स्पेशल), reph (र्ब), ब्ध cluster
P11 · pair F	EN · L2 public-health	Health camp notice (chemist)	Free Sugar Test Every Sunday / Shree Medical Store	Control for P12
P12 · pair F	HI · L2 public-health	Health camp notice (chemist)	हर रविवार मुफ्त शुगर जाँच / श्री मेडिकल स्टोर	Chandrabindu (ँ), nukta+half letter (फ़्त), थ्र; misinformation risk in a health notice
S13	MIX · L3 code-mixed	Festival offer (mobile shop)	Diwali Dhamaka Offer / हर मोबाइल पर मिफ्ट पक्का!	Two scripts on one poster, as real Indian ads do
S14	HI · L4 long	Subscription offer (tiffin service)	माँ के हाथ जैसा खाना / घर का बना शुद्ध शाकाहारी टिफ़िन / मासिक प्लान सिर्फ ₹2500 / फ़्री होम डिलीवरी	Length stress: 4 lines, ~17 words
S15	HI · L4 conjunct-dense	Festival greeting (Janmashtami)	श्री कृष्ण जन्माष्टमी की / हार्दिक शुभकामनाएँ	Densest conjuncts: थ्र, कृ (ऋ-matra), ष्ण, न्मा, ष्ट, दि (reph + i-matra)

ID	LANG · LEVEL	CATEGORY	EXACT TEXT REQUESTED	WHAT IT TESTS
S16	HI · L3 cultural grounding	Regional festival greeting (Chhath Puja)	छठ महापर्व की / हार्दिक शुभकामनाएँ	Under-represented regional festival: does the model know Chhath's visual vocabulary?

Full prompt, example P02. Every prompt is built from the same template; the exact text of each is stored in the generation log.

Design a square promotional poster for WhatsApp Status for Sharma Sweets, a family-run mithai shop in Kanpur, Uttar Pradesh. Show a brass thali of kaju katli and motichoor laddoos, lit clay diyas and a marigold garland. The poster must contain ONLY the following text, in Hindi (Devanagari script), spelled exactly as given:
 Line 1: "शुभ दीपावली"
 Line 2: "शर्मा स्वीट्स"
 Do not add any other words, letters, logos or watermarks. Style: bright, clean, professional design that a small local shop in India would proudly share with its customers.

B · Participant instructions and consent

Instructions shown to participants: Imagine you own a small shop in a Hindi-speaking town. You asked an AI tool to make a poster for your WhatsApp Status. On each screen you will see the exact text the shop owner asked for and three posters (A, B, C) from different AI models. The names are hidden and the order is shuffled. For each poster answer three quick questions, then pick the one you would post, or "None". Tap any image to zoom and check the text carefully.

Practice item: a mock poster reading "शुभ दीपावलि" against the requested "शुभ दीपावली". Correct answer: *small error* (short ि instead of long ी), with an explanation shown.

Consent (exact statement from the brief, a required checkbox): *"I confirm that I am 18 years or older. I voluntarily participated in this evaluation. I consent to my name, email, and responses/ratings being included in this assignment submission for hiring evaluation purposes."* The app stores the consent text and a timestamp with every rating; under-18s or no consent cannot continue.

Participants and consent record: PENDING - filled automatically from the rating files (name, email, age, Hindi level, consent text + timestamp per participant).

C · My audit notes per image

Per-image audit notes

PROMPT	MODEL	LANG	TEXT (MY LABEL)	NOTES
P01	GPT Image 1	EN	Exactly correct	All text correct; shop name set in caps
P02	GPT Image 1	HI	Small error	शुभ दीपावली correct; shop name rendered approx. 'शर्मा स्वीट्स' (doubled म, stray nukta under ट)
P03	GPT Image 1	EN	Exactly correct	Generic 'ATTA' packs, no brand
P04	GPT Image 1	HI	Exactly correct	All Hindi text correct; rice packs show fake 'AASHIRAAD' brand (Aashirvaad look-alike) - trademark risk
P05	GPT Image 1	EN	Exactly correct	₹ correct
P06	GPT Image 1	HI	Small error	सिर्फ़ rendered without nukta (सिर्फ); ब्लाउज़ and लक्ष्मी correct

PROMPT	MODEL	LANG	TEXT (MY LABEL)	NOTES
P07	GPT Image 1	EN	Exactly correct	All caps; phone number correct
P08	GPT Image 1	HI	Exactly correct	All 3 lines correct incl. विज्ञान, कक्षा, phone number
P09	GPT Image 1	EN	Exactly correct	Added unrequested line 'Agarwal General Store, Mathura' (legible English)
P10	GPT Image 1	HI	Major error	उपलब्ध rendered as 'उपल्वध'; added unrequested pseudo-Hindi line 'अगरावी गसरलातो सोरर' (garbled shop name)
P11	GPT Image 1	EN	Exactly correct	
P12	GPT Image 1	HI	Small error	मुफ्त without nukta; जाँच rendered as जांच (anusvara for chandrabindu)
S13	GPT Image 1	MIX	Major error	मोबाइल rendered as 'मोबाल'; added unrequested 'RAJ MOBILE POINT'; Apple logo on a phone
S14	GPT Image 1	HI	Major error	जैसा->'जेसा', डिलीवरी->'डिलेबरी', सिर्फ़/फ़्री lose nukta; टिफ़िन and शुद्ध correct
S15	GPT Image 1	HI	Exactly correct	Densest conjuncts (श्र, कृ, ष्ण, ष्ट, दि) all correct
S16	GPT Image 1	HI	Small error	शुभकामनाएँ rendered as 'शुभकामानाएँ' (extra ाँ); added unrequested English 'Maa Tara Fruit Bhandar'; Chhath visuals (soop, thekua, sugarcane, arghya at ghat) correct
P01	Gemini 2.5 Flash Image	EN	Exactly correct	All text correct
P02	Gemini 2.5 Flash Image	HI	Major error	दीपावली->'दोपावली'; shop name 'शमा स्कोटस' (शर्मा स्वीट्स)
P03	Gemini 2.5 Flash Image	EN	Exactly correct	Text correct; atta packs carry a garbled brand 'GUPIA ATTA'
P04	Gemini 2.5 Flash Image	HI	Major error	सभी->'सधी', छूट->'छाट', गुप्ता->'गुष्टा'; packs show garbled pseudo-brand labels
P05	Gemini 2.5 Flash Image	EN	Exactly correct	English text correct; added garbled mixed-script shop name 'लसमी टीलors'; real 'USHA' logo on sewing machine
P06	Gemini 2.5 Flash Image	HI	Gibberish / missing	'ब्लुज़ सेलही सफ़ि ₹250 / लस्मी टेलरस': every word misspelled, only ₹250 survives
P07	Gemini 2.5 Flash Image	EN	Exactly correct	Text correct; added 'Vidya Coaching Centre, Patna, Bihar' (legible)
P08	Gemini 2.5 Flash Image	HI	Major error	प्रारंभ->'प्रार्म', गणित और विज्ञान->'गगीत आरे विचान', संपर्क करें->'संपक कारें'; phone number correct
P09	Gemini 2.5 Flash Image	EN	Exactly correct	All text correct
P10	Gemini 2.5 Flash Image	HI	Major error	स्पेशल->'स्पषाल', उपलब्ध->'उजाब'; added garbled English 'MATHURA, UTTER UPREASSHI'
P11	Gemini 2.5 Flash Image	EN	Exactly correct	All text correct

PROMPT	MODEL	LANG	TEXT (MY LABEL)	NOTES
P12	Gemini 2.5 Flash Image	HI	Gibberish / missing	'हर रनिवार मुफिट शुमार जॉच्छ / श्री मडिजल स्तोष': every word wrong incl. the day of the week on a health notice
S13	Gemini 2.5 Flash Image	MIX	Major error	English line correct; मोबाइल->'मोबोल', गिफ्ट->'गिष्ट', पक्का->'पका'; added 'Raj Mobile Point' and garbled 'Meerut, Uprasshi'
S14	Gemini 2.5 Flash Image	HI	Gibberish / missing	Most words broken: 'हाठ', 'काना', 'गर', 'शूढ', 'टेप्नि', 'मासाक प्रान सीप्र', 'दलिरी'
S15	Gemini 2.5 Flash Image	HI	Gibberish / missing	'श्री कूश् जममताती की हादिक शुभफापाएँ': conjunct-dense line collapses
S16	Gemini 2.5 Flash Image	HI	Major error	महापर्व->'माहापर्व', हार्दिक->'हादिक', शुभकामनाएँ->'शुभाफामाएं'; added garbled Hindi sign 'मां तारा फ़ोट बन्दार, पतना, बेहार'; Chhath visuals (women in river at sunset, soop, thekua, sugarcane) correct
P01	Gemini 3.1 Flash Image Preview	EN	Exactly correct	All text correct
P02	Gemini 3.1 Flash Image Preview	HI	Exactly correct	शुभ दीपावली and शर्मा स्वीट्स both correct
P03	Gemini 3.1 Flash Image Preview	EN	Exactly correct	Text correct; generic 'WHEAT ATTA' packs
P04	Gemini 3.1 Flash Image Preview	HI	Exactly correct	All Hindi text correct; generic 'atta' packs
P05	Gemini 3.1 Flash Image Preview	EN	Exactly correct	All text correct
P06	Gemini 3.1 Flash Image Preview	HI	Exactly correct	All text correct incl. nukta in ब्लाउज़ and सिर्फ़ and लक्ष्मी; real 'USHA' logo on sewing machine
P07	Gemini 3.1 Flash Image Preview	EN	Exactly correct	All text correct
P08	Gemini 3.1 Flash Image Preview	HI	Exactly correct	All 3 lines correct; ॠ in विज्ञान drawn as a stylised ligature (to confirm with participants); English maths scribbles on the whiteboard background
P09	Gemini 3.1 Flash Image Preview	EN	Exactly correct	All text correct
P10	Gemini 3.1 Flash Image Preview	HI	Exactly correct	All text correct incl. उपलब्ध
P11	Gemini 3.1 Flash Image Preview	EN	Exactly correct	All text correct
P12	Gemini 3.1 Flash Image Preview	HI	Exactly correct	All text correct incl. मुफ़्त (nukta) and जौंच (chandrabindu)
S13	Gemini 3.1 Flash Image Preview	MIX	Exactly correct	Both lines correct; printed the instruction word 'WhatsApp Status' as a heading and added 'Raj Mobile Point'
S14	Gemini 3.1 Flash Image Preview	HI	Small error	All 4 lines correct except फ़्री rendered without nukta (फ्री)
S15	Gemini 3.1 Flash Image Preview	HI	Exactly correct	Densest conjuncts all correct

PROMPT	MODEL	LANG	TEXT (MY LABEL)	NOTES
S16	Gemini 3.1 Flash Image Preview	HI	Exactly correct	Text correct; most authentic Chhath scene (women in river at sunrise/sunset, soop with thekua, sugarcane, ghat)

D - Generation log

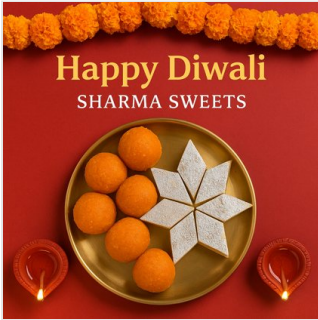
PROMPT	MODEL	PLATFORM	REQUEST ID	TIME (UTC)
P01	GPT Image 1	OpenAI Image API	req_ae419bb9c4e1493795ff53398e6d6946	2026-10-07 16:33
P02	GPT Image 1	OpenAI Image API	req_6187eea779d84054bbb5769d9e93ac3c	2026-10-07 16:34
P03	GPT Image 1	OpenAI Image API	req_1c22c369c3c94ed58d0352cf636d7d4b	2026-10-07 16:34
P04	GPT Image 1	OpenAI Image API	req_4559b11dd5334824901f3e3e1faa085e	2026-10-07 16:35
P05	GPT Image 1	OpenAI Image API	req_c44b42d1be854d669a0105a4b6d52147	2026-10-07 16:36
P06	GPT Image 1	OpenAI Image API	req_f45381e5d80d44cfac276c8c1cd3fbdd	2026-10-07 16:36
P07	GPT Image 1	OpenAI Image API	req_3e06700252db4ce49b5fa67ff14633e8	2026-10-07 16:37
P08	GPT Image 1	OpenAI Image API	req_3b78314b39e64612a6d6b67cb70716b7	2026-10-07 16:37
P09	GPT Image 1	OpenAI Image API	req_e0be16c55e06471f9f7a3bfe18fe2463	2026-10-07 16:38
P10	GPT Image 1	OpenAI Image API	req_eec1bdd071c94398a0b17b17e554ece7	2026-10-07 16:39
P11	GPT Image 1	OpenAI Image API	req_6af0392b0f394176ac7fdb392285722	2026-10-07 16:39
P12	GPT Image 1	OpenAI Image API	req_a9deaef23f7e4df0b108c2661c51513c	2026-10-07 16:40
S13	GPT Image 1	OpenAI Image API	req_2065d573b98942a0afd6b8380ec4bd92	2026-10-07 16:41
S14	GPT Image 1	OpenAI Image API	req_6a61951e925740b5a5c358269b5d7585	2026-10-07 16:41
S15	GPT Image 1	OpenAI Image API	req_e8ecde12c38f49228fb202af10684b93	2026-10-07 16:42
S16	GPT Image 1	OpenAI Image API	req_87183f30cb3a4b67b968e2ae2d55fa02	2026-10-07 16:43
P01	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	-YLGat3UMIrFg8UPiNSwoAQ	2026-10-07 17:35
P02	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	CoPGatHcOur1g8UPouSo4Qo	2026-10-07 17:36

PROMPT	MODEL	PLATFORM	REQUEST ID	TIME (UTC)
P03	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	FYPGaticG_mējuMPysDLyAQ	2026-10-07 17:36
P04	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	HoPGarfnNpTTg8UP7sy0A4	2026-10-07 17:36
P05	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	KIPGatSqBNy2g8UP6Nj0eA	2026-10-07 17:36
P06	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	M4PGapadGorFg8UPiNSwoAQ	2026-10-07 17:36
P07	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	04PGaoG-KK6lg8UPn5DrgQ8	2026-10-07 17:37
P08	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	RYPGaq_GGZusjuMPtN0mEQ	2026-10-07 17:37
P09	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	ToPGasia0YjWg8UP7afJoAY	2026-10-07 17:37
P10	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	WYPGavqfGruog8UP3ZnGmQQ	2026-10-07 17:37
P11	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	YoPGas2SMc0smNMPjfzR4AE	2026-10-07 17:37
P12	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	bIPGavn2LKrdg8UP7pTTiQQ	2026-10-07 17:37
S13	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	doPGatqtHobVqfkPwai9sA4	2026-10-07 17:38
S14	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	gIPGaq6N0rawg8UPiNXAkQQ	2026-10-07 17:38
S15	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	i4PGatSFEZTTg8UP7sy0A4	2026-10-07 17:38
S16	Gemini 2.5 Flash Image	Google Gemini API (generateContent)	lYPGaobuHPq2g8UPj9aJoAk	2026-10-07 17:38
P01	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	-oLGauDsB0HfG8UPw8X0gAQ	2026-10-07 17:35
P02	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	BYPGatnFIJjpjuMPq5-t2A4	2026-10-07 17:36
P03	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	V4PGav6vDojWg8UP7afJoAY	2026-10-07 17:37
P04	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	ZIPGapzmFriqg8UPq9ik0QQ	2026-10-07 17:37
P05	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	boPGav-jNN-ng8UPn5PY2A4	2026-10-07 17:37
P06	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	eYPGasTXF5HFg8UP1cG3mQQ	2026-10-07 17:38
P07	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	hIPGaoa3GffjjjuMP1tmXMQ	2026-10-07 17:38

PROMPT	MODEL	PLATFORM	REQUEST ID	TIME (UTC)
P08	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	j4PGaoy_JbjMqfkPnL-EuA4	2026-10-07 17:38
P09	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	nYPGauXXEL2iqfkP7L048Q8	2026-10-07 17:38
P10	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	p4PGasjSNZWng8UPyfu02AI	2026-10-07 17:38
P11	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	tIPGastACMDcg8UPk42_kAQ	2026-10-07 17:39
P12	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	vYPGaoLK CZ6kg8UPtbiQwA4	2026-10-07 17:39
S13	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	yIPGauz0P0f2juMPp8TSsAE	2026-10-07 17:39
S14	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	04PGatTzOMrBjuMP1aiEgAI	2026-10-07 17:39
S15	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	4oPGarWyE7rDjuMP-NP4EA	2026-10-07 17:39
S16	Gemini 3.1 Flash Image Preview	Google Gemini API (generateContent)	7YPGaojTN7nRg8UPqsnfiQQ	2026-10-07 17:40

Every record in `q1/generation_log.jsonl` also stores the exact prompt, settings and token usage.

E · All generated images

PROMPT	OPENAI · GPT IMAGE 1	GOOGLE · GEMINI 2.5 FLASH IMAGE	GOOGLE · GEMINI 3.1 FLASH IMAGE PREVIEW
P01 EN · Festival greeting (Diwali) Happy Diwali Sharma Sweets			
P02 HI · Festival greeting (Diwali) शुभ दीपावली शर्मा स्वीट्स			

P03

EN · Discount offer (kirana)
Flat 10% Off on All Groceries
Gupta Kirana Store



P04

HI · Discount offer (kirana)
सभी किराना सामान पर 10% की छूट
गुप्ता किराना स्टोर



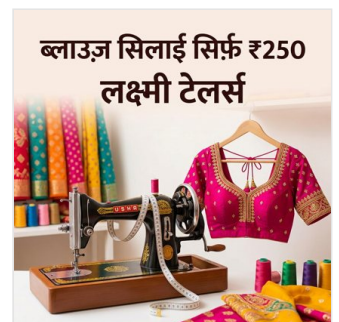
P05

EN · Service price (tailor)
Blouse Stitching Only ₹250
Laxmi Tailors



P06

HI · Service price (tailor)
ब्लाउज़ सिलाई सिर्फ ₹250
लक्ष्मी टेलर्स



P07

EN · Admissions notice (coaching)
Admissions Open
Classes 9 to 12: Maths and Science
Call: 98765 43210



P08

HI · Admissions notice
(coaching)
प्रवेश प्रारंभ
कक्षा 9 से 12: गणित और विज्ञान
संपर्क करें: 98765 43210



P09

EN · Festival product (Holi)
Holi Special
Herbal Colours Available



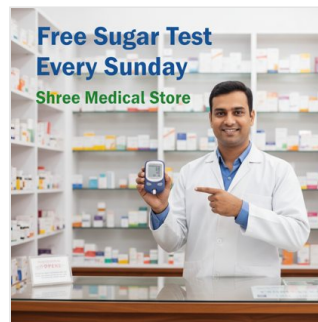
P10

HI · Festival product (Holi)
होली स्पेशल
हर्बल रंग उपलब्ध



P11

EN · Health camp notice
(chemist)
Free Sugar Test Every
Sunday
Shree Medical Store



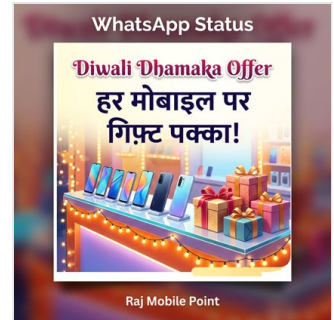
P12

HI · Health camp notice
(chemist)
हर रविवार मुफ्त शुगर जाँच
श्री मेडिकल स्टोर



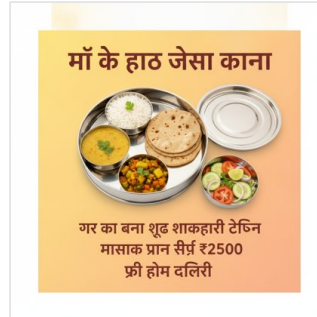
S13

MIX · Festival offer (mobile shop)
Diwali Dhamaka Offer
हर मोबाइल पर गिफ्ट पक्का!



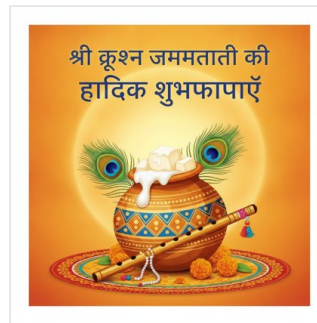
S14

HI · Subscription offer (tiffin service)
माँ के हाथ जैसा खाना
घर का बना शुद्ध शाकाहारी टिफिन
मासिक प्लान सिर्फ ₹2500
फ्री होम डिलीवरी



S15

HI · Festival greeting (Janmashtami)
श्री कृष्ण जन्माष्टमी की
हार्दिक शुभकामनाएँ



S16

HI · Regional festival greeting (Chhath Puja)
छठ महापर्व की
हार्दिक शुभकामनाएँ



Thumbnails shown; full-resolution PNGs are in [q1/images/](https://github.com/AbhyudayPatel/josh-talks-hindi-poster-eval) (<https://github.com/AbhyudayPatel/josh-talks-hindi-poster-eval>).

F · Q2 supporting tables

REAL DATA

H1. Signal validation against the crowd-consensus gold set (bad = CER > 15% vs medoid of ~126 transcriptions)

SIGNAL (GOLD SET, 5,544 TASKS)	TASKS FLAGGED	BAD IF FLAGGED (95% CI)	BAD OTHERWISE	NOTE
RTF < 1 (submitted faster than audio length)	273 (4.9%)	99.6% (98.0-99.9)	26.5%	
RTF < 0.5	163 (2.9%)	99.4% (96.6-99.9)	28.0%	

SIGNAL (GOLD SET, 5,544 TASKS)	TASKS FLAGGED	BAD IF FLAGGED (95% CI)	BAD OTHERWISE	NOTE
Unedited when Whisper was wrong	476 (8.6%)	100.0% (99.2-100.0)	23.5%	true by construction
Unedited (any clip)	476 (8.6%)	100.0% (99.2-100.0)	23.5%	true by construction (Whisper wrong on all gold clips)
RTF < 1 AND unedited	252 (4.5%)	100.0% (98.5-100.0)	26.7%	
RTF >= 1 AND unedited	212 (3.8%)	100.0% (98.2-100.0)	27.3%	true by construction
Blank on a speech clip	11 (0.2%)	100.0% (74.1-100.0)	29.9%	
Text density < 50% of consensus	205 (3.7%)	100.0% (98.2-100.0)	27.4%	

H2. Error rate by real-time factor (gold set)

RTF BIN	TASKS	BAD RATE	MEDIAN CER	UNEDITED SHARE
<0.5	163	99%	0.49	97%
0.5-1	110	100%	0.50	85%
1-2	178	82%	0.43	40%
2-4	755	35%	0.10	6%
4-8	1453	19%	0.06	1%
>8	2773	24%	0.07	3%

H3. Production tasks: RTF by clip length (8-digit task-ID block, real timings)

CLIP LENGTH	TASKS	MEDIAN RTF	SHARE RTF < 1	MEDIAN TIME (S)
<1s	12931	11.6	0.0%	4.4
1-3s	8502	5.2	4.9%	8.9
3-6s	7035	5.5	9.6%	23.5
6-10s	7861	4.2	16.8%	34.0
10-20s	12356	4.0	22.1%	47.3

H4. Per-user scorecard (10 users with history; timing-based fields omitted - synthetic)

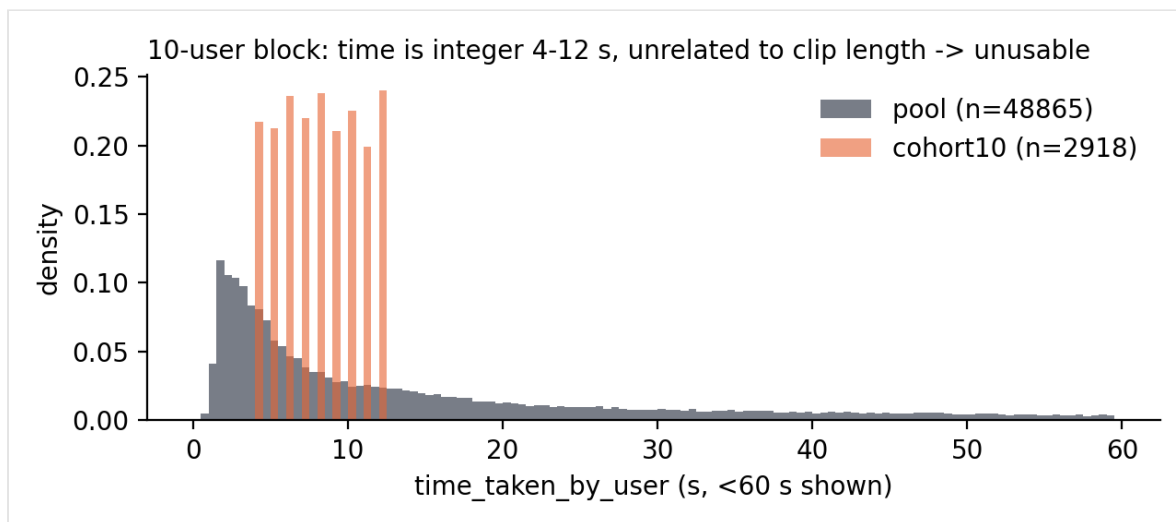
USER	TASKS	AUDIO MIN	UNEDITED (CI)	UNEDITED ≥5 S	MEDIAN EDIT WER	BLANK RATE	HALLUC. ACCEPTED/SEEN	JUNK	[BLANK]+TEXT	DUP. LONG TEXTS
632098	165	14	78% (68-85)	84%	0.27	2%	0/0	1	11	0
688393	86	7	36% (24-51)	48%	0.37	21%	0/2	0	6	0
580062	609	48	27% (22-32)	18%	0.25	17%	4/12	21	0	3
410566	349	28	21% (16-27)	17%	0.17	10%	2/5	0	15	0

USER	TASKS	AUDIO MIN	UNEDITED (CI)	UNEDITED ≥ 5 S	MEDIAN EDIT WER	BLANK RATE	HALLUC. ACCEPTED/SEEN	JUNK	[BLANK]+TEXT	DUP. LONG TEXTS
467270	176	14	14% (9-23)	11%	0.29	1%	0/4	0	0	1
393886	509	36	13% (10-18)	7%	0.17	3%	1/8	40	56	0
222099	333	27	7% (4-12)	3%	0.33	11%	0/1	5	0	1
681509	358	29	5% (3-9)	2%	0.31	8%	0/4	1	1	0
326611	149	11	5% (2-13)	0%	0.25	15%	0/0	0	0	0
311760	184	15	0% (0-4)	0%	0.16	12%	0/2	0	0	0

H5. Whisper hallucinations (strings recurring across unrelated recordings)

WHISPER STRING	TASKS	DISTINCT RECORDINGS	HUMANS WROTE [BLANK]	ACCEPTED VERBATIM
प्रस्तुति प्रस्तुति	398	338	62%	17%
झाल	134	122	68%	10%
कर दो	96	90	64%	19%
प्रतियों को प्रतियों को प्रतियों को प्रत...	10	10	40%	0%

H6. Why the 10-user timings are unusable



Reproducible: `q1/generate.py`, `q1/build_rating_app.py`, `q1/analyze.py`, `q2/q2_analysis.py`, `tools/build_submission.py`.